MAHESHBOOPATHI S

AI Engineer | Gen AI Specialist
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SUMMARY

AI Engineer with 2 years of experience building and deploying production-grade Agentic AI, RAG and LLM solutions across AWS and Azure. Reduced manual processing by 60% through MCP-based multi-agent workflows and improved model response accuracy by 40% with custom evaluation layers. Delivered a real-time voice sales solution using the OpenAI Realtime API and RAG, achieving 90%+ context-aware query accuracy, with hands-on experience across data ingestion, retrieval, deployment, and MLOps monitoring.

EXPERIENCE

AI Engineer
SetNext.ai — Bengaluru, India **Apr 2025 - Aug 2026**

- Developed an end-to-end multi-agent automation system using Model Context Protocol (MCP) to securely connect AI agents with enterprise tools, reducing manual processing workflows by 60%.
- Built a production-grade, real-time voice sales agent using the OpenAI Realtime API and Pinecone-backed RAG, achieving sub-second latency and 90%+ query accuracy.
- Developed a cloud-native LLM fine-tuning platform using AWS SageMaker and Azure ML, optimizing open-source models via LoRA/PEFT to boost response accuracy by 40%.
- Engineered an AI-driven Text-to-SQL interface using a semantic data model, enabling natural-language query retrieval from internal SAP databases.
- Built an automated Excel-to-SFTP secure ingestion pipeline, enabling seamless indexing and search capabilities across distributed enterprise datasets.

Software Trainee Intern
Aura Harks Technology — Chennai, India **Sep 2024 - Feb 2025**

- Developed an end-to-end predictive web application to analyze localized soil, weather, and agricultural metrics for regional farming.
- Trained, engineered, and evaluated SVM and Linear Regression models, achieving a stable 85% predictive accuracy on test datasets.
- Constructed an operational ML inference backend using Python and Flask, with MySQL managing structured environmental records.

KEY PROJECTS

Agentic AI Workflow Automation System

- Developed an end-to-end multi-agent automation system using Model Context Protocol (MCP) to securely connect AI agents with enterprise tools and data layers, reducing manual processing time by 60%.
- Built a high-precision Retrieval-Augmented Generation (RAG) pipeline utilizing Pinecone and specialized embeddings to achieve 90%+ context-aware query accuracy.
- Unified production AI model deployment and inference workflows across AWS and Microsoft Azure, leveraging AWS AI services and Azure AI Foundry to support scalable AI application development and deployment.

AI-Powered Real-Time Voice Sales Agent

- Delivered a production-ready real-time voice sales solution using the OpenAI Realtime API to enable low-latency, multi-turn customer conversations.
- Engineered real-time vector indexing and chunking pipelines using Pinecone-backed RAG to achieve 90%+ context-aware query accuracy during live conversations.

- Integrated tool-based sales workflows and automated follow-ups, with monitoring dashboards for tracking token usage, response latency, and user engagement.

LLM Fine-Tuning-as-a-Service Platform

- Developed a cloud-native platform for scalable LLM fine-tuning, evaluation, and secure deployment across AWS SageMaker and Azure Machine Learning.
- Optimized open-source LLMs using parameter-efficient fine-tuning techniques (LoRA, PEFT, and QLoRA), with automated REST APIs for dataset validation, version control, and multimodel pipeline tracking.
- Built systematic evaluation layers into inference pipelines, improving end-to-end model response accuracy by 40% while supporting multi-model pipeline tracking.

SKILLS

Generative & Agentic AI: LLMs, RAG, Prompt Engineering, Embeddings, LangChain, Multi-agent Systems, MCP Server Development

LLM Fine-Tuning: LoRA, PEFT, QLoRA, Instruction Tuning, Dataset Preparation

Speech & Voice AI: ASR, Whisper, OpenAI Realtime API

Programming Languages: Python, JavaScript, SQL, HTML, CSS

Backend & Messaging: FastAPI, Node.js, Redis

Deep Learning: PyTorch, Hugging Face Transformers

Cloud & AI Platforms: AWS SageMaker, Azure Machine Learning, Azure AI Foundry, Azure AI Services

Vector Databases: Pinecone, FAISS

MLOps & Tools: Docker, Git, CI/CD, MLflow

CERTIFICATIONS

Data Analytics with Python — NPTEL, IIT Roorkee

12-week program covering probability, hypothesis testing, regression, and clustering

EDUCATION

Master of Computer Applications (MCA)

Muthayammal Engineering College, Namakkal, Tamil Nadu | 2022 – 2024